

NESHAD D. PATHIRANA

Experimental Nuclear Physicist

Facility for Rare Isotope Beams, Michigan State University • East Lansing, Michigan, USA
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PROFESSIONAL SUMMARY

Ph.D. candidate at Michigan State University, based at the Facility for Rare Isotope Beams (FRIB), and former visiting scientist at the University of Oslo. Research focuses on experimental nuclear physics and its applications to nuclear structure, nuclear astrophysics, and neutrino physics. Employs precision measurements and advanced detector techniques to constrain key nuclear-structure properties, enabling indirect determinations of neutron-nucleus and neutrino-nucleus interactions with applications to nucleosynthesis in supernova environments and laboratory-based neutrino science.

Research Keywords: nuclear level densities; γ -ray strength functions; Gamow-Teller and Fermi transitions; single-particle and collective motions; charge-exchange and transfer reactions; β -decay; Oslo and Shape methods; multipole decomposition analysis; neutron-capture reactions; neutrino-nucleus reactions; supernova and neutrino simulations; magnetic spectrometers; silicon particle telescopes; γ -ray and neutron detectors.

EDUCATION

Doctor of Philosophy in Physics

2022 - Present

Facility for Rare Isotope Beams, Michigan State University | Michigan, USA

GPA: 4.00/4.00 | Three first-author papers

Master of Science in Physics

2022 - 2024

Facility for Rare Isotope Beams, Michigan State University | Michigan, USA

GPA: 4.00/4.00 | One first-author paper

Graduate Certificate in Instrumentation in High Energy Physics

2023 - 2024

Facility for Rare Isotope Beams, Michigan State University | Michigan, USA

GPA: 4.00/4.00

Bachelor of Science (Honors) in Physics

2017 - 2021

University of Peradeniya | Peradeniya, Sri Lanka

First Class Honors | GPA: 3.90/4.00 (Physics: 3.93; Mathematics: 4.00) | Top of the Class

RESEARCH EXPERIENCE

Graduate Research Assistant

May 2023 - Present

Facility for Rare Isotope Beams (FRIB), Michigan State University | Michigan, USA

- Participated in experiments using both forward and inverse kinematics at several nuclear physics facilities, including FRIB, RCNP, Argonne National Laboratory, and the Oslo Cyclotron Laboratory, employing a wide range of experimental configurations and detector systems for charged particles, neutrons, and γ -rays
- Developed the Charge-Exchange Oslo method using data collected with the S800 magnetic spectrograph in coincidence with the GRETINA high-purity germanium γ -ray tracking array at NSCL
- Extracted charged-current neutrino-nucleus reactions relevant to supernova nucleosynthesis and neutrino-detector applications using high-resolution charge-exchange data obtained with the Grand Raiden magnetic spectrometer at RCNP
- Performed theoretical reaction calculations, and astrophysical and neutrino simulations
- Contributed to the upgrade of the Low Energy Neutron Detector Array (LENDa) by testing and evaluating novel scintillator materials to improve neutron detection performance
- Lead and contribute to the submission of experimental proposals for the FRIB Program Advisory Committee (PAC)

Visiting Scientist

Apr - Jul 2025

Norwegian Nuclear Research Center (NNRC), University of Oslo | Oslo, Norway

- Joined a research team at NNRC, gaining experience with forward kinematics light-ion beams, hands-on operation of detector arrays, and online and offline data acquisition systems at the Oslo Cyclotron Laboratory
- Experimentally constrained a reaction important for understanding early Solar System formation, using high resolution transfer reaction data obtained with the SiRi particle-telescope and the OSCAR LaBr₃ detector array at the Oslo Cyclotron Laboratory
- Evaluated the impact of the experimentally constrained reaction rate in supernova environments and its Galactic chemical evolution

Visiting Collaborator*Research Center for Nuclear Physics (RCNP), Osaka University | Osaka, Japan***October - November 2025**

- Joined a research team at RCNP, gaining experience with forward kinematics light-ion beams and hands-on operation of the Grand Raiden spectrometer, SAKRA silicon particle telescopes, LaBr₃ detector arrays, and online and offline data acquisition systems

*ATLAS User Facility, Argonne National Laboratory (ANL) | Illinois, USA***November 2024 | July 2026**

- Joined research teams at ANL, gaining experience with proton and nuCARIBU beams, hands-on operation of the GRETINA and SuN++ detector arrays, and online and offline data acquisition systems

Undergraduate Research Assistant**Jul 2019 - Jun 2021***University of Peradeniya | Peradeniya, Sri Lanka*

- Designed and constructed a portable cylindrical ionization chamber, differential-amplifier electronics, and Arduino-based signal processing for laboratory identification of ionizing particles
- Performed an independent study of electromagnetic and hadronic calorimeters used in particle detectors

Junior Research Assistant**May - Sep 2016***National Institute of Fundamental Studies | Kandy, Sri Lanka*

- Supported projects in quantum to classical transitions, quantum chaos, quantum computing and nonlocality, and low-cost brain computer interface hardware and software

TEACHING EXPERIENCE**Graduate Teaching Assistant****Aug 2022 - Aug 2023***Michigan State University | East Lansing, MI, USA*

- Instructor for Introductory Physics Laboratory II (PHY 252), and Visions of the Laboratory (ISP 205L) for 200 level science and non-science major students: prepared and evaluated laboratory handouts; supervised experiments
- Tutorial instructor for Mathematical Modeling in Physics (PHY 415) for 400 level physics honors students: graded tutorials and conducted discussion sessions

Graduate Teaching Assistant**Aug 2021 - Jul 2022***University of Peradeniya | Peradeniya, Sri Lanka*

- Instructor for nuclear physics laboratory sessions for M.Sc. in physics students, and advanced undergraduate physics laboratory sessions for 300/400 level students enrolled in the physics honors degree program: prepared and evaluated laboratory handouts; supervised experiments
- Instructor for Quantum Mechanics I (PH 430), Quantum Mechanics II (PH 431), and General Relativity (PH 458) for 400 level physics honors students: prepared marking schemes, graded tutorials and conducted discussion sessions
- Conducted help desk sessions for Quantum Mechanics and Atomic Physics (PH 230) for 200 level students

AWARDS, FELLOWSHIPS & HONORS**2025-2026** Norwegian Nuclear Research Center (NNRC) Fellowships (\$14,000)**2025** Galonsky International Travel Award (\$3500)**2025-2024** FRIB Fellowships (\$2250)**2025-2023** IReNA Travel Grants (\$16,000)**2024** Graduate School Travel Fellowship (\$600)**2024** Herbert T. Graham Scholarship (\$550)**2022** Prof. Lakshman Dissanayake Gold Medal for Excellence in Physics**2022** Prof. George Dissanaikie Memorial Gold Medal in Physics**2022** Dr. C. A. Hevavitharana Memorial Prize in Physics**2022** Prof. A. W. Wolfendale Prize in Physics**2022** University of Peradeniya Award for Academic Excellence**2021** Merit Award, SURS Symposium**2018** Eramudugolla Dunuwila Prize in Mathematics**Honor societies:** Elected member: Phi Kappa Phi Honor Society and Sigma Xi Scientific Research Honor Society**CONFERENCES, WORKSHOPS, AND SUMMER SCHOOLS****Invited Oral Presentations:**

2026 Invited plenary: "Development of the Charge-Exchange Oslo method for experiments with stable and unstable beams", *FRIB-TA Topical Program: Statistical Nuclear Properties in the AI/ML Era: Advances, Challenges, and Synergies for FRIB Science*, FRIB, MI, USA

2026 Invited plenary: "Development of the Charge-Exchange Oslo method and Application Towards Constraining Reaction Rates for Nucleosynthesis of Cosmochronometer ⁹²Nb", *18th International Symposium on Capture Gamma-Ray Spectroscopy and Related Topics (CGS18)*, Monterey, CA, USA

2025 Invited seminar: “Development of the Charge-Exchange Oslo method and applications toward constraining reaction rates for ^{92}Nb nucleosynthesis”, *Department of Physics Seminar Series, University of Oslo, Oslo, Norway*

2024 Invited seminar: “Career Opportunities with a Physics Degree”, *Department of Physics Seminar Series, University of Peradeniya, Sri Lanka*

Contributed Oral Presentations:

2026 (upcoming) “Development of the Charge-Exchange Oslo method and applications toward constraining reaction rates for ^{92}Nb nucleosynthesis”, *12th Nuclear Physics in Astrophysics (NPAXII) International Conference, Cluj-Napoca, Romania*

2026 “Oslo studies near ^{92}Nb and the implications for stellar nucleosynthesis”, *10th Workshop on Nuclear Level Density and Gamma Strength, University of Oslo, Norway*

2025 “Development of the Charge-Exchange Oslo method and applications toward constraining reaction rates for ^{92}Nb nucleosynthesis”, *Research Presentation Seminar, Facility for Rare Isotope Beams, East Lansing, MI, USA*

2025 “Solving the puzzle of the cosmochronometer ^{92}Nb production sites”, *1st s-, i-, and r-Element Nucleosynthesis Conference (sirEN), Giulianova, Italy*

2024 “Development of the Charge-Exchange Oslo method and applications toward constraining reaction rates for ^{92}Nb nucleosynthesis”, *8th p-Process Workshop, Konkoly Observatory, Budapest, Hungary*

2024 “Development of the Charge-Exchange Oslo method and applications toward constraining reaction rates for ^{92}Nb nucleosynthesis”, *Research Presentation Seminar, Facility for Rare Isotope Beams, East Lansing, MI, USA*

2021 “Construction of an ionization chamber to detect alpha and beta particles”, *RESCON Research Congress, University of Peradeniya, Sri Lanka*

2019 “Use of Electromagnetic & Hadronic Calorimeters in Particle Detection”, *Department of Physics, University of Peradeniya, Sri Lanka*

Contributed Poster Presentations:

2025 “Solving the puzzle of the cosmochronometer ^{92}Nb production sites”, *1st s-, i-, and r-Element Nucleosynthesis Conference (sirEN), Giulianova, Italy*

2024 “Development of the Charge-Exchange Oslo method and the first application to constrain (n,γ) cross sections”, *11th Nuclear Physics in Astrophysics (NPAXI) International Conference, Dresden, Germany*

2024 “First application of the Charge-Exchange Oslo method to constrain (n,γ) cross sections”, *21st Exotic Beam Summer School, Argonne National Laboratory, IL, USA*

2024 “First application of the Charge-Exchange Oslo method to constrain (n,γ) cross sections”, *9th Workshop on Nuclear Level Density and Gamma Strength, University of Oslo, Norway*

2023 “Charge-exchange reactions in conjunction with the Oslo method”, *1st IReNA–Ukakuren Conference, NAOJ, Tokyo, Japan*

2021 “Construction of an ionization chamber to detect alpha and beta particles”, *SURS Undergraduate Research Symposium, University of Peradeniya, Sri Lanka*

Other Attended Conferences:

2025 “Mini-symposium on Precision Measurements at Low Energies”, *FRIB, MI, USA*

2023 “IReNA Workshop on Weak Interactions in Nuclear Astrophysics”, *FRIB, MI, USA*

2023 “CeNAM Frontiers in Nuclear Astrophysics Meeting”, *FRIB, MI, USA*

Participated Summer Schools:

2025 “FRIB Theory Alliance Summer School: Emergence of Collective Motion in Atomic Nuclei”, *FRIB, MI, USA*

2025 “Open Questions and Research Tools in Nuclear Astrophysics: Summer School and Hackathon”, *Beaver Island, MI, USA*

2024 “11th Nuclear Physics in Astrophysics International Summer School”, *Stolpen, Germany*

2024 “21st Exotic Beam Summer School”, *Argonne National Laboratory, IL, USA*

2023 “IReNA Frontiers in Nuclear Astrophysics Summer School”, *FRIB, MI, USA*

TECHNICAL SKILLS

Nuclear Physics Tools: Oslo Method software; Shape Method software; TALYS; FOLD

Astrophysical and Neutrino Tools: NuGrid, Winnet, and MESA reaction network calculations; MARLEY

Data Acquisition Tools: FRIB, OCL, RCNP, and ANL offline/online data acquisition and analysis systems

Experimental Systems used during Ph.D. Research: S800 and Grand Raiden Spectrographs; GRETINA (HPGe) and OSCAR (LaBr₃) γ-ray detector arrays; SiRi particle telescope; LENDA neutron detector, AT-TPC detector, SuN++ detector, Liquid-Hydrogen target systems

Programming and Scientific Computing: C++; ROOT; Python; PAW; Geant4; LISE++; MATLAB; Wolfram Mathematica; Origin; LaTeX

PROFESSIONAL AFFILIATIONS & SERVICE

- Member: American Physical Society (APS)
- Member: NuGrid Collaboration

- Member: International Research Network for Nuclear Astrophysics (IReNA), JINA Center for the Evolution of the Elements (JINA-CEE), and Center for Nuclear Astrophysics Across Messengers (CeNAM)
- Former Secretary: Sri Lankan Student Association at Michigan State University
- Former Secretary: Physical Society, University of Peradeniya
- Former Member: Musical Society, Astronomical Society, Mathematical Society, and Music Society at University of Peradeniya
- Former Leader: High School Science and Astrophysics Quiz Teams

PUBLICATIONS

First-author Publications:

1. **N. D. Pathirana**, R. G. T. Zegers, et al., “*Extraction of neutron-capture cross sections on ^{92}Zr using the charge-exchange Oslo method.*” *Physical Review C* 113, 015801 (2026). [DOI](#)
2. **N. D. Pathirana**, M. Markova, A.C. Larsen, M. Pignatari, et al., “*Enhanced supernova production of ^{92}Nb and its implications for Galactic Chemical Evolution.*” Under review at Nature Communications; Research Square preprint (2026). [Preprint](#)
3. **N. D. Pathirana**, R. G. T. Zegers, et al., “*Experimental study of supernova neutrino-induced nucleosynthesis of ^{92}Nb .*” Manuscript in preparation
4. **N. D. Pathirana**, R. G. T. Zegers, et al., “*Charged-current neutrino- ^{208}Pb cross sections via the $^{208}\text{Pb}(^3\text{He},t)$ charge-exchange reaction.*” Manuscript in preparation
5. **N. D. Pathirana** and D. R. A. Mendis “*Construction of an ionization chamber to detect alpha and beta particles.*” Research Congress (RESCON 2021), Postgraduate Institute of Science, University of Peradeniya (2021). [Abstract](#)

Co-author Publications:

1. C. O’Shea, G. Lotay, A. Gade, D. T. Doherty, H. Schatz, J. S. Randhawa, **N. D. Pathirana** et al., “*Single-nucleon transfer on unstable ^{59}Cu probes the NiCu cycle in astrophysical X-ray bursts.*” *Physical Review Letters* (2026). [DOI](#)
2. L. A. Riley, I. Conroy, A. M. Himmelreich, M. Heinze, J. Kosa, B. McNulty, **N. D. Pathirana** et al., “*Determination of proton and neutron contributions to the $0^+_{g.s.} \rightarrow 2^+_1$ excitations in ^{42}Si and ^{44}S using inelastic proton scattering in inverse kinematics and intermediate-energy Coulomb excitation.*” *Physical Review C* 112, 014331 (2025). [DOI](#)

REFERENCES

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